

$$a) 3\sec^2(x) - 7 = -1$$

$$3\sec^2(x) = 6$$

$$\sec^2(x) = 2$$

$$\frac{1}{\cos^2(x)} = 2$$

$$\cos^2(x) = \frac{1}{2}$$

$$\cos(x) = \pm \frac{\sqrt{2}}{2}$$

$$\cos(x) = \frac{\sqrt{2}}{2}$$

$$\text{or } \cos(x) = -\frac{\sqrt{2}}{2}$$

$$x = \frac{\pi}{4}, 2\pi - \frac{\pi}{4} = \frac{7\pi}{4}$$

$$x = \pi - \frac{\pi}{4} \text{ or } \pi + \frac{\pi}{4}$$

$$= \frac{3\pi}{4}$$

$$= \frac{5\pi}{4}$$

$$\therefore x = \frac{\pi}{4}, \frac{7\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}$$

$$b) 1 + \sin(x) = \sqrt{3} \cos(x)$$

$$[1 + \sin(x)]^2 = [\sqrt{3} \cos(x)]^2$$

$$1 + 2\sin(x) + \sin^2(x) = 3\cos^2(x)$$

$$1 + 2\sin(x) + \sin^2(x) = 3(1 - \sin^2(x))$$

$$1 + 2\sin(x) + \sin^2(x) - 3 + 3\sin^2(x) = 0$$

$$4\sin^2(x) + 2\sin(x) - 2 = 0$$

$$2(2\sin^2(x) + \sin(x) - 1) = 0$$

$$2((2\sin(x) - 1)(\sin(x) + 1)) = 0$$

$$\sin(x) = \frac{1}{2} \text{ or } \sin(x) = -1$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \quad x = \frac{3\pi}{2}$$

$$\therefore x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

$$C) \cos^2(2x) = 3\sin^2(x) - 2$$

$$(1 - 2\sin^2(x))^2 = 3\sin^2(x) - 2$$

$$1 - 4\sin^2(x) + 4\sin^4(x) = 3\sin^2(x) - 2$$

$$4\sin^4(x) - 7\sin^2(x) + 3 = 0$$

$$(4\sin^2(x) - 3)(\sin^2(x) - 1) = 0$$

$$\sin^2(x) = \frac{3}{4} \text{ or } \sin^2(x) = 1$$

$$\sin(x) = \pm \frac{\sqrt{3}}{2} \text{ or } \sin(x) = \pm 1$$

$$\sin(x) = \frac{\sqrt{3}}{2} \text{ or } \sin(x) = -\frac{\sqrt{3}}{2}$$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}$$

$$x = \frac{4\pi}{3}, \frac{5\pi}{3}$$

$$\sin(x) = 1 \text{ or } \sin(x) = -1$$

$$x = \frac{\pi}{2}$$

$$x = \frac{3\pi}{2}$$

$$\therefore x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{2}$$

$$d) \frac{\cos(2x)}{\cos(x) + \sin(x)} = 1$$

$$\frac{\cos^2(x) - \sin^2(x)}{\cos(x) + \sin(x)} = 1$$

$$\frac{[\cos(x) - \sin(x)][\cos(x) + \sin(x)]}{\cos(x) + \sin(x)} = 1$$

$$[\cos(x) - \sin(x)][\cos(x) + \sin(x)] = \cos(x) + \sin(x)$$

$$[\cos(x) - \sin(x)][\cos(x) + \sin(x)] - (\cos(x) + \sin(x)) = 0$$

$$[\cos(x) - \sin(x)][\cos(x) + \sin(x) - 1] = 0$$

$$\cos(x) - \sin(x) = 0 \quad \text{or} \quad \cos(x) + \sin(x) - 1 = 0$$

$$\cos(x) = \sin(x)$$

$$\cos(x) + \sin(x) = 1$$

$$[\cos(x) + \sin(x)]^2 = 1^2$$

$$\cos^2(x) + 2\cos(x)\sin(x) + \sin^2(x) = 1$$

$$\sin(2x) + 1 = 1$$

$$\sin(2x) = 0$$

$$2x = 0, \pi, 2\pi, 3\pi, 4\pi$$

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi$$

divide both
sides by
 $\cos(x)$

$$1 = \tan(x)$$

$$x = \frac{\pi}{4}, \frac{5\pi}{4}$$